**Master's Degree in Cognitive Science**

NEURAL MASS AND SPIKING MODELS FOR THE STUDY OF NEURAL CIRCUITS

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# Contents

# Abstract

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# Chapter 1

# Stochastic Resonance in a Spiking Neural Network Model of the Honeybee Antennal Lobe

## 1.1 Introduction

One of the first well established findings in neuroscience with the advent of electrical recordings of neuronal activity, was the variability of neurons' responses. That is, when stimulated with the same input, neurons respond with different temporal patterns of spikes, while the overall firing rate remains roughly equal [1–3].

This trial-to-trial variability is commonly called noise. Sources of noise in neurons are many and not all variability in response is due to noise (see [4] for a review). Still, due to the physical scale of cells, a stochastic component in neuronal function is inevitable by thermal noise alone. It is then to be expected that neural systems would have evolved to tolerate or, more daringly, exploit the presence of noise.

### 1.1.1 Classical Stochastic Resonance

The idea of studying the effects of noise on the dynamics of a system first appeared parallely in the work of Benzi et al. [5] and Nicolis [6], where the term Stochastic Resonance (SR) was used to explain the periodic behavior of earth's average temperature.

A brief covering of that first SR application will be made, as it provides a practical understanding of the general phenomenon and a surprising link to neural systems.

The most common way to model temperature averages was using simple energy balance deterministic models:

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = R_{in}(T) - R_{out}(T) \tag{1.1}$$

where $C$ is the earth's thermal capacity, and $R_{in}$ and $R_{out}$ represent the total incoming and outgoing solar radiation, defined as

$$\begin{aligned} R_{in} &= Q\mu \\ R_{out} &= \alpha(T)Q\mu + \varepsilon(T) \end{aligned}$$

where $\mu$ is the average incoming solar radiation.

In [5] a simplification is used: 3 steady-states are assumed to exist as $T_1 < T_2 < T_3$, those are possible earth's climates. 1.1 then becomes

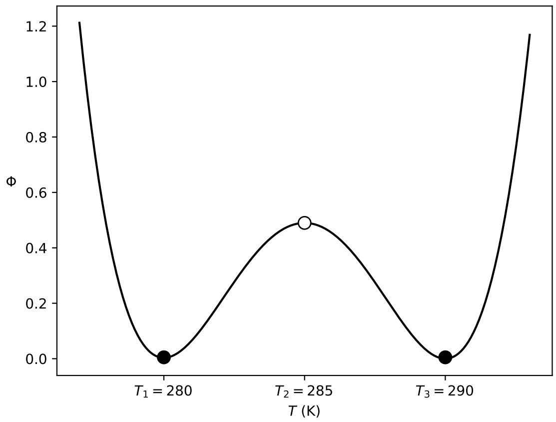


Figure 1.1: The double well potential. Filled circles are stable points.

$$C\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\mu\varepsilon(T)}{1+\beta\left(1-T/T_1\right)\left(1-T/T_2\right)\left(1-T/T_3\right)} - \varepsilon(T) \tag{1.2}$$

To analyze the stability of the defined climates the rate of change of temperatures is taken as

$$F(T) = \frac{\mathrm{d}T}{\mathrm{d}t}\ , \tag{1.3}$$

then a pseudo-potential energy function is defined as

$$\Phi(T) = -\int F(T)\mathrm{d}T \tag{1.4}$$

that represents the "energy" needed to change the temperature across temperature values. From $\Phi$ we get

$$F(T) = -\frac{\partial\Phi}{\partial T} \tag{1.5}$$

Then intuitively the maximum of $\Phi$ will be an unstable point, while the minima will be stable. That is, where the second derivative with respect to temperature

$$\frac{\mathrm{d}F}{\mathrm{d}T} = -\frac{\partial^2\Phi}{\partial T^2} \tag{1.6}$$

is $< 0$, any neighboring starting state of the earth's temperature will converge to it, while when it is $> 0$ they will diverge.

It can be shown that $T_1$, $T_3$ are stable points, while $T_2$ is unstable: a peak any starting state would "roll off" from (see A). In our context, $T_1$, $T_3$ are the only observable climates, temperatures at which the earth will stay for enough time to be detectable in the geological records. A system where 2 stable states separated by an unstable state exist is called a bistable system, and the plot of $\Phi$ over $T$ displays the typical double well shape of the (pseudo)potential function (Figure 1.1).

With the present model, we can see that the earth's climate would depend solely (all else being equal) on the starting temperature, i.e., wether it is smaller or greater than $T_2$. Then it would follow $F(T)$ towards $T_1$ or $T_3$ (Figure 1.2). However, earth's temperatures do indeed show oscillations with a period of $10^5$ years.

A weak cosinusoidal forcing modulating the input solar radiations can be introduced, representing the varying distance of the earth from the sun along its elliptic orbit. Then $\mu$ becomes time dependent

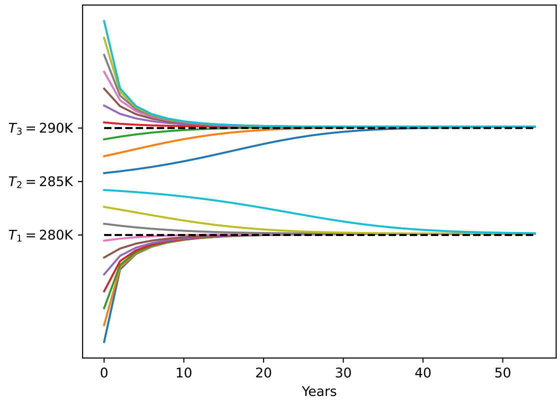


Figure 1.2: Trajectories for different starting temperatures without noise.

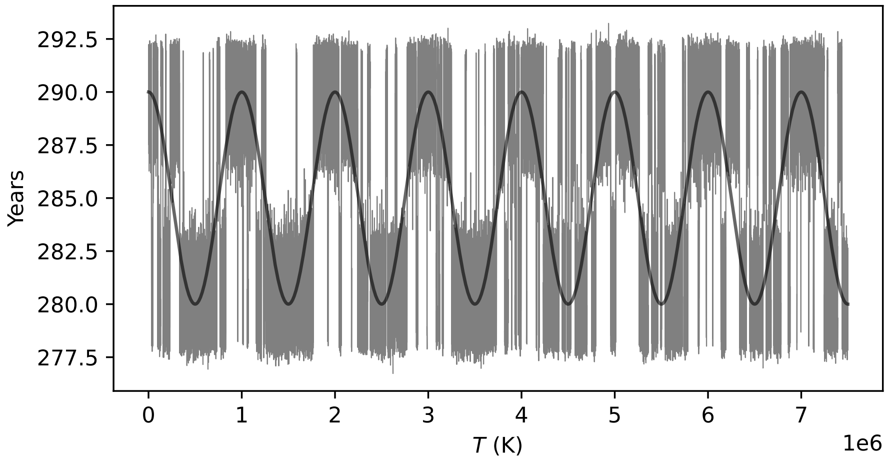


Figure 1.3: Temperature over time from numerical solution of 1.2, with $\sigma = 0.15$.

$$\mu(t) = 1 + \gamma \cos(\omega t) \tag{1.7}$$

where $\gamma \ll 1$ and $\omega = 2\pi/10^5$, reflecting the observed oscillation period. The derivative and pseudo-energy function become time dependent $\tilde{F} = \tilde{F}(T;t)$, $\tilde{\Phi} = \tilde{\Phi}(T;t)$.

Still, this periodicity alone does not lead to any change in climates: the system never jumps over $T_2$ (see fig).

If gaussian white noise, representing various mechanisms causing $T$ fluctuations at a smaller time scale, is added to the system, yielding

$$\frac{\mathrm{d}T}{\mathrm{d}t} = F(T) + \sigma\eta(t) \tag{1.8}$$

then for certain values of $\sigma$ a jump between stable states becomes possible, with a frequency $\approx 2\pi/10^5$. That is, the system shows temperature oscillations at the forcing frequency, as noise allows the system to jump from one equilibrium to another (Figure 1.3). Note that oscillations arise from the joint effect of the periodic forcing and noise: noise pushes the system away from the equilibrium at every timestep, but only when the forcing asymmetrically tilts the double well potential such that the distance between the current equilibrium and the unstable one becomes small enough to allow the stochastic component to push the system over it, a climate transition (that is, a jump from $T_1$ to $T_3$ or viceversa) can occur. This can be seen as an ”escape” from a stable equilibrium, requiring a jump over a (pseudo)energy potential ”wall”. This is a classic problem in physics, Kramer’s rate [7] gives the frequency of jumps per second

$$r_K = \frac{\sqrt{V''(x_m)/m}\sqrt{|V''(x_b)/m|}}{2\pi\gamma}e^{-\frac{\Delta V}{D}} \tag{1.9}$$

where $V''(x_m)$ is the curvature around the minima, $V''(x_b)$ that at the maximum, $\Delta V$ the distance from the minimum to the maximum, and $D = \sigma^2/2$ the temperature of the system (noise intensity in our context). The inverse $\frac{1}{r_K}$ gives the residence time, the time spent at one of the minima. If our condition for SR is that it allows the system to follow the subthreshold forcing, then the optimal $D$ is such that, given the forcing period $T_\omega$ and the residence time as a function of $D$, $T_K(D) = \frac{1}{r_K(D)}$ is

$$T_K(D) = \frac{T_\omega}{2} \tag{1.10}$$

Some useful considerations can be made on SR defined on the climate model. First, it requires a system containing a nonlinearity, as in 1.2, such that 2 stable states separeted by a potential wall exist. Additionally, the periodic forcing must be insufficient alone to cause a crossing (subthreshold). Later it will be shown that these assumptions can be relaxed. Last, as 1.10 suggests, too low $D$ will not be able to push the system over the barrier at the frequency needed, and high enough $D$ make the potential barrier effectively nonexistant, increasing $r_K$ and making the double well model meaningless.

Hence, given a metric of how well the system follows the forcing, there will be a peak for a certain $D$. In this sense, there is a resonance in the system. For classic SR models, the natural measure is the spectral power (obtained by Fourier transform of the system response) at the forcing frequency relative to the neighboring ones: Signal-to-Noise Ratio (SNR) [8].

### 1.1.2 Small detour on dynamical system theory

It will be shown that neurons are better modeled by 2D systems, due to the coexistence and interaction of 2 main ionic currents that govern a neuron’s state $\left(\mathrm{Na}^+ \text{ and } \mathrm{K}^+\right)$, that determine a depolarization-hyperpolarization feedback loop. To analyze a 2+D systems the tools of dynamical systems theory are indispensable. A brief covering of the main methods will follow, as they will be referenced in the following chapters (for a more complete tractation, see [9]).

For any 2D system defined by

$$\dot{v} = f\,(v, u) \tag{1.11}$$

$$\dot{u} = g\,(v, u) \tag{1.12}$$

the system exists in the space $S = \{v\mathbf{e}_1 + u\mathbf{e}_2 \mid v, u \in \mathbb{R}\}$, and is always identified with the state vector $\mathbf{s} = (v, u)$. Then we can define the mapping $F : \mathbb{R}^2 \to \mathbb{R}^2$ such that $F(\mathbf{s}) \mapsto (f(\mathbf{s}), g(\mathbf{s}))$, the standard definition of a vector field (in $\mathbb{R}^n$). Then the direction of $F(\mathbf{s})$ will be given by the sign of $f(\mathbf{s})$ and $g(\mathbf{s})$. So the curves defined by $f(v, u) = 0$ and $g(v, u) = 0$, the $v$- and $u$-nullclines, that separate the plane in changes in the direction of $\dot{v}$ and $\dot{u}$, are of particular interest. Note that $\dot{v} = \mathrm{d}v(t)/\mathrm{d}t$ and $\dot{u} = \mathrm{d}u(t)/\mathrm{d}t$. Then the evolution of the system’s state from an initial position $\mathbf{s}(t_0) = (v(t_0), u(t_0)) = \mathbf{s}_0$ up to a time $t$, $\mathbf{s}(t)$, is governed by the flow of the vector field, formally $\varphi_t : S \to S$. Indeed, the trajectory $\phi_{t-t_0}(s_0)$ is a curve always tangent to $F$.

All points of intersection of the nullclines are such that $\dot{v} = \dot{u} = 0$. Consider the state $\mathbf{s}^*$ such that $f(\mathbf{s}^*) = g(\mathbf{s}^*) = 0$, if the system remains at $\mathbf{s}^*$ for all $t \geq 0$ then $\mathbf{s}^*$ is an equilibrium. Equilibria can be stable or unstable, the stability of an equilibria can be probed by analyzing the response of the system at $\mathbf{s}^*$ to a small disturbance $\mathbf{s}_d$. From to the time view, we are interested in $\varphi_{t_0-t}(\mathbf{s}_d - \mathbf{s}^*)$, for some time $t \geq 0$. However, the trajectory based on $f$, $g$ usually cannot be calculated when they are nonlinear (as is the case for 1.16). Then we use the linearization of the system near the equilibria, in linear algebra terms solving

$$\dot{\mathbf{s}} = \mathrm{A}\mathbf{s} \equiv \begin{pmatrix} \dot{v} \\ \dot{u} \end{pmatrix} = \underbrace{\begin{pmatrix} \frac{\partial f(\mathbf{s}^*)}{\partial v} & \frac{\partial f(\mathbf{s}^*)}{\partial u} \\ \frac{\partial g(\mathbf{s}^*)}{\partial v} & \frac{\partial g(\mathbf{s}^*)}{\partial u} \end{pmatrix}}_{\mathrm{J}} \begin{pmatrix} v \\ u \end{pmatrix} \tag{1.13}$$

note that in the Jacobian matrix quadratic and higher orders terms are omitted since they are evaluated on the small disturbance $\mathbf{s}_d - \mathbf{s}^*$, and so are assumed to contribute extremely little to the solution.

To solve the system in 1.13 we can use the eigenvalue method (if the eigenvectors are linearly independent) where a general solution is $\mathbf{s}(t) = e^{\lambda t}\boldsymbol{\xi}$, where $\boldsymbol{\xi}$ is the eigenvector. substituting into 1.13 we get

$$\dot{\mathbf{s}} = \mathrm{A}\mathbf{s} \rightarrow \mathrm{A}\boldsymbol{\xi} = \lambda\boldsymbol{\xi} \tag{1.14}$$

The eigenvalues of A are found solving the characteristic equation $\det(\mathrm{A} - \lambda \mathrm{I}) = 0$

Knowing equilibria, their type and stability, joined with knowing $F$, provides information on all possible behaviors of a system. This can all be visualized plotting the equilibria and $F$, along with the nullclines, creating a *phase portait*.

### 1.1.3 From Earth to neurons

Historically the most used (and first) model to study SR was that of 1.2, more generally expressed as a stochastic differential equation [10]

$$\frac{\mathrm{d}x}{\mathrm{d}t} = x\left(a - bx^2\right) + \mu\gamma\cos(\omega t) + \sigma\xi(t) \tag{1.15}$$

where $\xi$ is gaussian white noise, and $x(a - bx^2)$ cubic term, that for $a > 0$ has 3 fixed points. When defining the pseudo-potential as in 1.4 it gives the quartic polynomial that is the simplest realization of the double potential well (here ignoring forcing and noise component)

$$\begin{aligned} \Phi &= -\int -x^3 + ax \\ &= \frac{b}{4}x^4 - \frac{a}{2}x^2 + C \end{aligned}$$

Intuitively, a neuron is a system that displays both quiescent and spiking behavior: that is, 2 stable states, separated by a "barrier" commonly called the voltage threshold. Note that the term threshold is hardly accurate: different current values can all elicit spikes in neurons depending on stimulation duration, some neurons respond to certain stimulation frequencies at currents lower than those needed to elicit a spike with a pulse-like stimulation, and others fire in response to hyperpolarizing currents (known as a rebound spike) [11, Chapter 7].

Unsurprisingly, a neuron can be modeled in the same form of 1.15 to show SR [12]. Interestingly, the same holds in more accurate neuron models.

The FitzHugh-Nagumo (FHN) model [13, 14] is a 2D reduction of the Hodgkin-Huxley (HH) biophysical model [15], its attractivness stems from being able to recreate the spiking behavior of neurons without the 4 differential equations that regulate the HH neuron, allowing for easier analysis of the model using the theory of dynamical systems.

The FH model is defined by a fast and a slow variable

$$\frac{\mathrm{d}v}{\mathrm{d}t} = v\,(a - v)\,(v - 1) - w + I \tag{1.16}$$

$$\frac{\mathrm{d}w}{\mathrm{d}t} = bv - cw \tag{1.17}$$

with $v$ representing the (fast) membrane voltage and $w$ the (slow) feedback recovery mechanisms of the membrane (e.g., opening of $K^+$ channels). The main difference of this model with 1.2 is the second equation for $w$ that allows for an additional behavior that cannot be seen in the climate model. However, a cubic term for $\dot{v}$ is present and one may wonder wether the resulting double well potential is a requirement for SR. It will be shown that it is not, and that it would describe an unrealistic regime for neural function.

# Appendix A

# Stability of Equilibria

In Benzi et al. [5]

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